

Tutorial – Week 11

Black-box Testing

Question 1

Consider a software application that calculates the monthly payments of a car loan. The application accepts three inputs:

- Cost of the car: The cost should be a positive integer between \$1,000 and \$200,000.
 - Loan period: The period of the loan in years should be a positive integer between 1 and 10 years.
 - Interest rate: The annual interest rate should be a positive decimal between 0.01 (1%) and 0.2 (20%).
- a. Create Normal Boundary Value cases to test the functionality of this application.
 - b. Create Robust Boundary Value cases to test the functionality of this application.
 - c. Create Worst-case Boundary Value cases to test the functionality of this application.
 - d. Create Robust Worst-case Boundary Value cases to test the functionality of this application.

Answer:

Boundary values

- Cost of the car: \$1000 (min), \$1001 (min+), \$100000(nom), \$199999 (max-), and \$200,000 (max).
- Loan period: 1 year (min), 2 years (min+), 5 years (nom), 9 years (max-), and 10 years (max).
- Interest rate: 0.01 (min), 0.02 (min+), 0.1 (nom), 0.19 (max-), and 0.2 (max).

Invalid values

- Cost of the car: Lower invalid value \$999, Upper invalid value \$200,001.
- Loan period: Lower invalid value 0 years, Upper invalid value 11 years.
- Interest rate: Lower invalid value 0, Upper invalid value 0.21.

a. Normal Boundary Value cases

Number of test cases = $4(n) + 1 = 4(3) + 1 = 13$ test cases.

Case	Cost of the car	Load period	Interest rate	Expected result
1	\$100000	5	0.01	
2	\$100000	5	0.02	
3	\$100000	5	0.1	
4	\$100000	5	0.19	
5	\$100000	5	0.2	

6	\$100000	1	0.1	
7	\$100000	2	0.1	
8	\$100000	9	0.1	
9	\$100000	10	0.1	
10	\$1000	5	0.1	
11	\$1001	5	0.1	
12	\$199999	5	0.1	
13	\$200000	5	0.1	

b. Robust Boundary Value cases

Number of test cases = $6(n) + 1 = 6(3) + 1 = 19$ test cases.

Case	Cost of the car	Load period	Interest rate	Expected result
1	\$100000	5	0	
2	\$100000	5	0.01	
3	\$100000	5	0.02	
4	\$100000	5	0.1	
5	\$100000	5	0.19	
6	\$100000	5	0.2	
7	\$100000	5	0.21	
8	\$100000	0	0.1	
9	\$100000	1	0.1	
10	\$100000	2	0.1	
11	\$100000	9	0.1	
12	\$100000	10	0.1	
13	\$100000	11	0.1	
14	\$999	5	0.1	
15	\$1000	5	0.1	
16	\$1001	5	0.1	
17	\$199999	5	0.1	
18	\$200000	5	0.1	
19	\$200001	5	0.1	

c. Worst-case Boundary Value cases

Number of test cases = Cartesian product the possible values for 3 variables = $5^3 = 125$ test cases.

Case	Cost of the car	Load period	Interest rate	Expected result
1	\$1000	1	0.01	
2	\$1000	1	0.02	
	\$1000	1	0.1	
	\$1000	1	0.19	
	\$1000	1	0.2	

	\$1000	2	0.01	
	\$1000	2	0.02	
	\$1000	2	0.1	
	\$1000	2	0.19	
	\$1000	2	0.2	
	\$1000	5	0.2	
	\$1000	5	0.01	
	\$1000	5	0.02	
	\$1000	5	0.1	
	\$1000	5	0.19	
	\$1000	5	0.2	

d. Robust Worst-case Boundary Value cases

Number of test cases = Cartesian product the possible values for 3 variables = $7^3 = 343$ test cases.

Question 2

Consider a function $f(x1, x2)$ where the values of $x1$ and $x2$ are defined to be $1000 \leq x1 \leq 10000$ and $10\text{-year} \leq x2 \leq 50\text{-year}$.

Assume the equivalence classes for $x1$ and $x2$ are: -

V1 = [1000 to 5000]

V2 = [5001 to 10000]

V3 = [10-year to 19-year]

V4 = [20-year to 29-year]

V5 = [30-year to 50-year]

- Using graphical representation, show the minimum test cases one could generate using weak normal equivalence class testing.
- Why a strong normal equivalence class testing generates more test cases compare to a weak normal equivalence class testing?
- List the equivalence classes of $x1$ and $x2$ for robustness testing.
- Using graphical representation, show the minimum test cases you could generate using the weak robust equivalence class testing.

Question 3

Consider an online e-commerce platform that calculates shipping costs based on three factors: the type of item being shipped (Fragile or Non-Fragile), the chosen shipping method (Standard, Express, or Overnight), and the total order price (Below \$50, \$50-\$100, Above \$100).

The rules for shipping costs are as follows:

- Standard shipping costs \$5 for Non-Fragile items and \$10 for Fragile items.
- Express shipping costs \$10 for Non-Fragile items and \$15 for Fragile items.
- Overnight shipping costs \$20 for Non-Fragile items and \$25 for Fragile items.
- Orders below \$50 incur an additional handling fee of \$5, regardless of item type or shipping method.
- Orders between \$50 and \$100 have no additional fees.
- Orders above \$100 get free shipping, regardless of item type or shipping method.

Create a decision table to represent the logic of this system and describe how you would convert this decision table into test cases for the e-commerce platform.

Answer

	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
Item Type	Fragile	Fragile	Fragile	Non-Fragile	Non-Fragile	Non-Fragile	-
Shipping Method	Standard	Exp	ON	Standard	Exp	ON	-
Order Price	<\$50	<\$50	<\$50	<\$50	<\$50	<\$50	>\$100
Shipping Fee	\$10	\$15	\$25	\$5	\$10	\$20	\$0
Handling Fee	\$5	\$5	\$5	\$5	\$5	\$5	\$0
Total	\$15	\$20	\$30	\$10	\$15	\$25	

	Case 8	Case 9	Case 10	Case 11	Case 12	Case 13
Item Type	Fragile	Fragile	Fragile	Non-Fragile	Non-Fragile	Non-Fragile
Shipping Method	Standard	Exp	ON	Standard	Exp	ON
Order Price	\$50-100	\$50-100	\$50-100	\$50-100	\$50-100	\$50-100
Shipping Fee	\$10	\$15	\$25	\$5	\$10	\$20
Handling Fee	\$0	\$0	\$0	\$0	\$0	\$0
Total	\$10	\$15	\$25	\$5	\$10	\$20